MURANG'A UNIVERSITY OF TECHNOLOGY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

UNIVERSITY ORDINARY EXAMINATION

2024/2025 ACADEMIC YEAR

THIRD YEAR **FIRST** SEMESTER EXAMINATION FOR BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING

SCS 302– ARTIFICIAL INTELLIGENCE

DURATION: 2 HOURS

INSTRUCTIONS TO CANDIDATES:

1. Answer question ONE and any other two questions.
2. Mobile phones are not allowed in the examination room.
3. You are not allowed to write on this examination question paper.

SECTION A – ANSWER ALL QUESTIONS IN THIS SECTION

QUESTION ONE (30 MARKS)

(a) Using your own words give a difinition of ‘’Artificial Intelligence’’ (2 marks)

(b) Discuss **THREE** objectives of studying artificial intelligence. (3 marks)

(c) Explain the meaning of the concept ‘’knowledge representation’’ (2 marks)

(d) Answer the following questions by indication whether the statements are true (T) or (F)

(i) Depth search is often slower than breadth search (1 mark)

(ii) Draughts and scrabble are both deterministic games (1 mark)

(e) Outline **FOUR** differences between an agent and other software. (4 marks)

(f) Explain the meaning of the following concept:

i. Abductive reasoning (2 marks)

ii. Case based reasoning (2 marks)

iii. Plausible reasoning (2 marks)

(g) Using examples explain **TWO** types of mobile agents (2 marks)

(h) Explain FOUR types of agents environments (4 marks)

(i) Use fourth table to determine the following propositional calculus sentences

i. (TP^Q) VR: where P and R have truth value F, AND Q has truth value T. (2 marks)

ii. T(P^Q)^(TPVR): where P, Q and R have truth value Γ (3 marks

SECTION B – ANSWER ANY TWO QUESTIONS IN THIS SECTION

QUESTION TWO (20 MARKS)

(a) Explain the differences between mini and alpha beta algorithms. (4 marks)

(b) Describe FOUR types of board games that can be implemented using artificial intelligence tasks. (4 marks)

(c) Distinguish between a ‘senter’ and an ‘actator’ in the context of intelligent agent technology. (4marks)

(d) Discuss **THREE** main difference between supersized and unsupervised learning. (4 marks)

(e) Differentiate between classification and clustering in machine learning. (4 marks)

QUESTION THREE (20 MARKS)

(a) Discuss the meaning of the concept ‘’Influence engine’’ (2 marks)

(b) Discuss the elements of a search problem in the context artificial intelligence. (5 marks)

(c) Explain **THREE** limitations of using rules as a technique for representing knowledge. (3 marks)

(d) Explain **FOUR** characteristics of a good knowledge representation technique. (4 marks

(e) Explain the meaning of the following concepts:

i. Intrapersonal intelligence (2 marks)

ii. Backward charming (2 marks)

iii. Forward charming (2 marks)

QUESTION FOUR (20 MARKS)

(a) Explain the meaning of the term ‘knowledge system’ (2 marks)

(b) Draw a well labelled structure of an expert system (4 marks)

(c) Explain the function of each part of an expert system drawn in (b) (7 marks)

(d) Describe FOUR benefits of including an explanation on facility in an expert system. (4 marks)

(e) Explain any THREE applications of knowledge systems (3marks)